

# Wipers and Washers



| Special Tool(s)                                                                                |                                                                                                                                                |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>ST2574-A   | Flex Probe Kit<br>NUD105-R025D or equivalent                                                                                                   |
| <br>ST3093-A   | Fluke 77-IV Digital Multimeter<br>FLU77-4 or equivalent                                                                                        |
| <br>ST2834-A  | Vehicle Communication Module (VCM)<br>and Integrated Diagnostic System (IDS)<br>software with appropriate hardware, or<br>equivalent scan tool |
| <br>ST3094-A | Test Lamp SGT27000 or equivalent 250-<br>350 mA incandescent bulb test lamp<br>SGT27000                                                        |

## Principles of Operation

### Windshield Wiper Operation

The wiper system is activated by the multifunction switch. The multifunction switch is mounted to the Steering Column Control Module (SCCM) . The SCCM commands the wiper motor to activate the appropriate wash, low, high and intermittent modes through a private Local Interconnect Network (LIN) .

When the multifunction switch is in the HIGH position, the wiper motor also receives a ground signal which is hardwired directly through the SCCM (bypassing the logic within the module). This fail-safe circuit makes sure that the wipers function in HIGH speed mode in the event of a LIN communication failure.

Vehicles equipped with a front video camera have a front camera washer control switch that is part of the multifunction switch.

### Washer System (if not equipped with front video camera)

When the multifunction switch is placed in the WASH position, the SCCM sends the request to the windshield wiper motor through the LIN . The windshield wiper motor supplies voltage to the washer pump, which then directs washer fluid to the windshield.

### Washer System (if equipped with front video camera)

When the multifunction switch is placed in the WASH position, the windshield wiper motor supplies voltage to the washer pump. Ground is provided to the washer pump through the normally closed contact in the front camera washer relay (located in the auxiliary Battery Junction Box (BJB) ).

When the front video camera washer control is activated, the **SCCM** sends the request to the windshield wiper motor through the **LIN** . The windshield wiper motor controls the direction of the washer pump to direct washer fluid to the front video camera.

### **Courtesy Wipe Feature**

When the washer system is activated, the wipers cycle 3 times with washer fluid, followed by one final wipe a few seconds later to clear any remaining fluid that is dripping from the top of the windshield. This feature can be enabled/disabled through the message center. For information on enabling/disabling this feature, refer to the Owner's Literature.

### **Speed Dependent Wipers**

If equipped, the speed dependent feature increases the frequency of wiping to compensate for the extra moisture that accumulates on the windshield as the vehicle speed increases. This feature activates when the multifunction switch is in any intermittent setting except the INT 1 position. As vehicle speed increases, the speed dependent feature gradually shortens the delay between wipes. The **SCCM** receives Vehicle Speed Sensor (VSS) information from the Body Control Module (BCM) through the High Speed Controller Area Network (HS-CAN) . The **SCCM** sends the **VSS** information to the wiper motor through the **LIN** .

### **Rain Sensitive Wipers (if equipped)**

When equipped, the rain sensitive wiper feature replaces the intermittent wiper feature. The rain sensitive feature is active when the multifunction switch is in any of the 5 auto/intermittent sensitivity settings. The setting closest to OFF is the least sensitive to moisture, the highest setting being the most sensitive to moisture.

The rain sensor optically senses water on the windshield. An infrared beam of specific intensity lights a section of the windshield. The intensity of the reflected beam is measured and compared to the intensity of the transmitted beam. If there is water on the surface of the windshield, the beam distorts, reducing the intensity of the beam. If the beam is not reflected completely, it is interpreted as rain on the windshield and the windshield wipers are started.

The rain sensor sends data through the **LIN** to the windshield wiper motor. When the rain sensitive wiper feature is enabled, based on the data received from the rain sensor and the auto/intermittent setting on the multifunction switch, the system automatically activates and adjusts the wiper speed and frequency when moisture is detected on the windshield.

A gel-type lens is used as the interface between the rain sensor and the windshield. If the lens is missing or damaged, the rain sensing wipers can be inoperative. The rain sensor gel-type lens is part of the rain sensor and cannot be replaced separately on a rain sensor of this type.

The rain sensitive wiper feature can be enabled/disabled through the message center.

When the rain sensitive wipers are disabled from the message center, or the windshield wiper motor detects a fault with the rain sensor input, the windshield wiper motor changes to a default intermittent mode. The timing of the wipers correspond to a standard intermittent setting based on the auto/intermittent sensitivity setting on the multifunction switch.

### **Wiper Activated Headlamps**

If equipped with automatic headlamps, the **SCCM** sends a message to the **BCM** over the **HS-CAN** to illuminate the exterior lamps within 10 seconds of the wipers being activated by the multifunction switch when the headlamp control switch is in the AUTOLAMP position. The **BCM** continues to illuminate the exterior lamps for approximately 30 seconds after the wipers have been turned off.

If equipped with rain sensitive wipers (feature must be enabled through the message center) and the multifunction switch is in 1 of the 5 auto/intermittent settings, the **SCCM** sends a message to the **BCM** over the **HS-CAN** to illuminate the exterior lamps within 30 seconds of the wipers being activated by moisture detected on the windshield when the headlamp control switch is in the AUTOLAMP position. The **BCM** continues to illuminate the exterior lamps for approximately 60 seconds after the wipers have been deactivated (moisture no longer detected on the windshield).

Inspection and Verification

- 1. Verify the customer concern.
- 2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

| Mechanical                                                                                                                                   | Electrical                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Wiper linkage</li><li>• Washer nozzles</li><li>• Washer fluid hoses</li><li>• Wiper blades</li></ul> | <ul style="list-style-type: none"><li>• Battery Junction Box (BJB) fuse(s):<ul style="list-style-type: none"><li>▪ 41 (15A) (vehicles equipped with front video camera only)</li><li>▪ 49 (30A)</li><li>▪ 54 (5A) (vehicles equipped with front video camera only)</li><li>▪ 79 (5A)</li></ul></li><li>• Wiper relay</li><li>• Rain sensor (if equipped)</li><li>• Wiper motor</li><li>• Washer pump</li><li>• Wiring, terminals or connectors</li></ul> |

- 3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to next section.
- 4. **NOTE:** *Make sure to use the latest scan tool software release.*  
  
If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .
- 5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*  
  
If the scan tool does not communicate with the VCM :
  - check the VCM connection to the vehicle.
  - check the scan tool connection to the VCM .
  - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.
- 6. If the scan tool does not communicate with the vehicle:
  - verify the ignition is ON.
  - verify the scan tool operation with a known good vehicle.
  - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.
- 7. Carry out the network test.
  - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
  - If the network test passes, retrieve and record the Continuous Memory Diagnostic Trouble Codes (CMDTCs) .
- 8. Clear the CMDTCs and carry out the self-test diagnostics for the Steering Column Control Module (SCCM) and the Body Control Module (BCM) .
- 9. If the DTCs retrieved are related to the concern, refer to the Steering Column Control Module (SCCM) DTC Chart. For all other SCCM DTCs, refer to Section 211-05. For any BCM DTCs, refer to Section 419-10.
- 10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

**NOTE:** This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. Continuous Memory Diagnostic Trouble Codes (CMDTCs) have an additional 2-character DTC status code suffix to assist in determining DTC history.

Steering Column Control Module (SCCM) DTC Chart

| DTC            | Description                                      | Action                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B1008:09       | Wiper Mode Switch: Component Failure             | <ul style="list-style-type: none"><li>▪ If the windshield wipers are inoperative, INSTALL a new multifunction switch. REFER to Section 211-05. CLEAR the DTCs and REPEAT the self-test. If the DTC is retrieved again, INSTALL a new SCCM . REFER to Section 211-05.</li><li>▪ If the windshield wipers are on continuously, <a href="#">GO to Pinpoint Test C.</a></li></ul> |
| B1051:09       | Front Washer Switch: Component Failure           | INSTALL a new multifunction switch. REFER to Section 211-05. CLEAR the DTCs and REPEAT the self-test. If the DTC is retrieved again, INSTALL a new SCCM . REFER to Section 211-05.                                                                                                                                                                                            |
| B10AD:02       | Rain Sensor: General Signal Failure              | <a href="#">GO to Pinpoint Test E.</a>                                                                                                                                                                                                                                                                                                                                        |
| B10AD:08       | Rain Sensor: Bus Signal / Message Failure        | <a href="#">GO to Pinpoint Test E.</a>                                                                                                                                                                                                                                                                                                                                        |
| B10AD:49       | Rain Sensor: Internal Electronic Failure         | INSTALL a new rain sensor. REFER to Rain Sensor in this section. CLEAR the DTCs. REPEAT the self-test.                                                                                                                                                                                                                                                                        |
| B1131:02       | Wiper Motor Module: General Signal Failure       | <a href="#">GO to Pinpoint Test B.</a>                                                                                                                                                                                                                                                                                                                                        |
| B1131:08       | Wiper Motor Module: Bus Signal / Message Failure | <a href="#">GO to Pinpoint Test B.</a>                                                                                                                                                                                                                                                                                                                                        |
| B12F7:09       | Single Wipe Switch: Component Failure            | INSTALL a new multifunction switch. REFER to Section 211-05. CLEAR the DTCs and REPEAT the self-test. If the DTC is retrieved again, INSTALL a new SCCM . REFER to Section 211-05.                                                                                                                                                                                            |
| All other DTCs | —                                                | REFER to Section 211-05.                                                                                                                                                                                                                                                                                                                                                      |

Symptom Chart

| Symptom Chart                                                                                |                                                                                                                                                       |                                                                                          |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Condition                                                                                    | Possible Causes                                                                                                                                       | Action                                                                                   |
| <ul style="list-style-type: none"><li>• All windshield wiper modes are inoperative</li></ul> | <ul style="list-style-type: none"><li>• Fuse</li><li>• Wiring, terminals or connectors</li><li>• Wiper relay</li><li>• Multifunction switch</li></ul> | <ul style="list-style-type: none"><li>• <a href="#">GO to Pinpoint Test A.</a></li></ul> |

|                                                                                                                           |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                           | <ul style="list-style-type: none"> <li>Wiper motor</li> <li>Steering Column Control Module (SCCM)</li> </ul>                                                                                                                                                                                  |                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>The intermittent and/or low speed windshield wipers are inoperative</li> </ul>     | <ul style="list-style-type: none"> <li>Wiring, terminals or connectors</li> <li>Multifunction switch</li> <li>Wiper motor</li> <li>Rain sensor (if equipped)</li> <li>SCCM</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li><a href="#">GO to Pinpoint Test B.</a></li> </ul>                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>The high speed windshield wipers are inoperative</li> </ul>                        | <ul style="list-style-type: none"> <li>Windshield wiper motor</li> </ul>                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>INSTALL a new windshield wiper motor. REFER to Wiper Motor.</li> </ul>                                                                                                                                              |
| <ul style="list-style-type: none"> <li>The windshield wipers stay on continuously</li> </ul>                              | <ul style="list-style-type: none"> <li>Wiring, terminals or connectors</li> <li>Wiper relay</li> <li>Multifunction switch</li> <li>Wiper motor</li> <li>SCCM</li> </ul>                                                                                                                       | <ul style="list-style-type: none"> <li><a href="#">GO to Pinpoint Test C.</a></li> </ul>                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>The windshield wipers do not operate correctly in the intermittent mode</li> </ul> | <ul style="list-style-type: none"> <li>Multifunction switch</li> <li>Wiper motor</li> <li>SCCM</li> </ul>                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>If the vehicle is not equipped with rain sensitive wipers, <a href="#">GO to Pinpoint Test D.</a></li> <li>If the vehicle is equipped with rain sensitive wipers, <a href="#">GO to Pinpoint Test E.</a></li> </ul> |
| <ul style="list-style-type: none"> <li>The rain sensitive wiper feature is inoperative</li> </ul>                         | <ul style="list-style-type: none"> <li>Fuse</li> <li>Wiring, terminals or connectors</li> <li>Rain sensitive feature not enabled</li> <li>Foreign material on windshield, such as dirt, dust or bugs</li> <li>Wiper blades</li> <li>Windshield glass</li> <li>Multifunction switch</li> </ul> | <ul style="list-style-type: none"> <li><a href="#">GO to Pinpoint Test E.</a></li> </ul>                                                                                                                                                                   |

|                                                                                                                                                 |                                                                                                                                                                                                                                   |                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Wiper motor</li> <li>• Rain sensor</li> <li>• SCCM</li> </ul>                                                                                                                            |                                                                                            |
| <ul style="list-style-type: none"> <li>• The rain sensitive wipers have erratic/inconsistent operation</li> </ul>                               | <ul style="list-style-type: none"> <li>• Foreign material on windshield, such as dirt, dust or bugs</li> <li>• Wiper blades</li> <li>• Windshield glass</li> <li>• Rain sensor</li> </ul>                                         | <ul style="list-style-type: none"> <li>• <a href="#">GO to Pinpoint Test E.</a></li> </ul> |
| <ul style="list-style-type: none"> <li>• The rain sensitive wipers are too sensitive/not sensitive enough to moisture</li> </ul>                | <ul style="list-style-type: none"> <li>• Foreign material on windshield, such as dirt, dust or bugs</li> <li>• Wiper blades</li> <li>• Damaged windshield glass</li> <li>• Multifunction switch</li> <li>• Rain sensor</li> </ul> | <ul style="list-style-type: none"> <li>• <a href="#">GO to Pinpoint Test E.</a></li> </ul> |
| <ul style="list-style-type: none"> <li>• The windshield wipers do not park at the correct position</li> </ul>                                   | <ul style="list-style-type: none"> <li>• Wiper mounting arm and pivot shaft</li> <li>• Wiper motor</li> </ul>                                                                                                                     | <ul style="list-style-type: none"> <li>• <a href="#">GO to Pinpoint Test F.</a></li> </ul> |
| <ul style="list-style-type: none"> <li>• The washer pump is inoperative or always on - vehicles not equipped with front video camera</li> </ul> | <ul style="list-style-type: none"> <li>• Wiring, terminals or connectors</li> <li>• Washer pump</li> <li>• Multifunction switch</li> <li>• Wiper motor</li> <li>• SCCM</li> </ul>                                                 | <ul style="list-style-type: none"> <li>• <a href="#">GO to Pinpoint Test G.</a></li> </ul> |
| <ul style="list-style-type: none"> <li>• The washer pump is inoperative or always on - vehicles equipped with front video camera</li> </ul>     | <ul style="list-style-type: none"> <li>• Wiring, terminals or connectors</li> <li>• Washer pump</li> <li>• Multifunction switch</li> <li>• Wiper motor</li> <li>• Front camera washer relay</li> </ul>                            | <ul style="list-style-type: none"> <li>• <a href="#">GO to Pinpoint Test H.</a></li> </ul> |

Pinpoint Test A: The Windshield Wipers Are Inoperative

Refer to Wiring Diagrams Cell [81](#) , Wipers and Washers for schematic and connector information.

Normal Operation

The wiper relay (located in the Battery Junction Box (BJB) ) is energized when the ignition is ON. The wiper relay provides voltage to the wiper motor. The multifunction switch is mounted to the Steering Column Control Module (SCCM) . The SCCM commands the wiper motor to activate the appropriate wash, low, high and intermittent modes through a Local Interconnect Network (LIN) .

When the multifunction switch is in the HIGH position, the wiper motor receives ground which is hardwired directly through the SCCM (bypassing the logic within the SCCM and the wiper motor). This fail-safe circuit makes sure that the wipers function in HIGH speed mode in the event of a LIN communication failure.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Wiper relay
- Multifunction switch
- Wiper motor
- SCCM

PINPOINT TEST A : ALL WINDSHIELD WIPER MODES ARE INOPERATIVE

**NOTICE:** Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK THE WIPER RELAY

- Disconnect: Wiper Relay .
- Substitute a known good relay and recheck the windshield wiper motor operation.

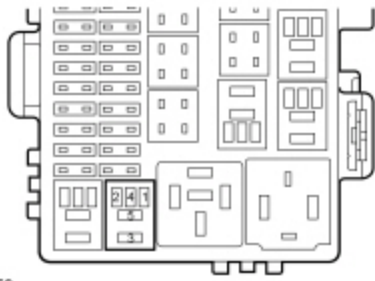
Do the wipers operate?

|     |                                                         |
|-----|---------------------------------------------------------|
| Yes | REMOVE the known good relay. INSTALL a new wiper relay. |
| No  | REMOVE the known good relay. GO to <a href="#">A2</a> . |

A2 CHECK THE BATTERY SUPPLIED VOLTAGE CIRCUIT TO THE WIPER RELAY FOR AN OPEN

- Ignition OFF.
- Measure the **voltage** between:

| Positive Lead     |            | Negative Lead |         |
|-------------------|------------|---------------|---------|
| Pin               | Circuit    | Pin           | Circuit |
| Wiper relay pin 3 | SBB49 (RD) | —             | Ground  |



• N0136159

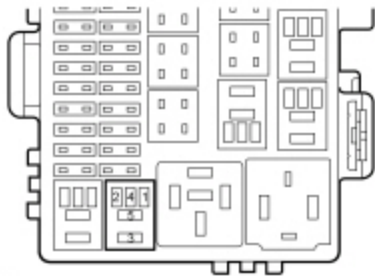
Is the voltage greater than 11 volts?

|     |                                                                                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | GO to <a href="#">A3</a> .                                                                                                                                                 |
| No  | VERIFY the <u>BJB</u> fuse 49 (30A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

**A3 CHECK THE IGNITION SUPPLIED VOLTAGE CIRCUIT TO THE WIPER RELAY FOR AN OPEN**

- Ignition ON.
- Measure the **voltage** between:

| Positive Lead     |               | Negative Lead |         |
|-------------------|---------------|---------------|---------|
| Pin               | Circuit       | Pin           | Circuit |
| Wiper relay pin 2 | CDC64 (WH/BU) | —             | Ground  |



• N0136159

Is the voltage greater than 11 volts?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">A4</a> . |
| No  | REPAIR the circuit.        |

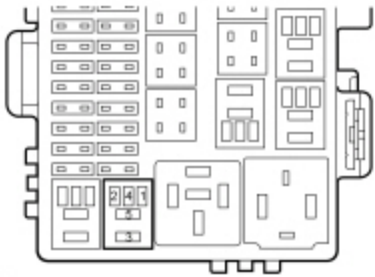
**A4 CHECK THE WIPER RELAY COIL GROUND CIRCUIT FOR AN OPEN**

- Measure the **voltage** between:

| Positive Lead | Negative Lead |
|---------------|---------------|
|---------------|---------------|



|                   |               |                   |               |
|-------------------|---------------|-------------------|---------------|
| Pin               | Circuit       | Pin               | Circuit       |
| Wiper relay pin 2 | CDC64 (WH/BU) | Wiper relay pin 1 | GD123 (BK/GY) |



N0136159

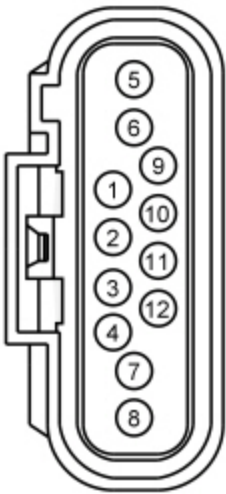
Is the voltage greater than 11 volts?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">A5</a> . |
| No  | REPAIR the circuit.        |

A5 CHECK THE WINDSHIELD WIPER MOTOR VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Connect: Wiper Relay .
- Disconnect: Windshield Wiper Motor [C125](#).
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead              |               | Negative Lead |         |
|----------------------------|---------------|---------------|---------|
| Pin                        | Circuit       | Pin           | Circuit |
| <a href="#">C125</a> Pin 8 | CRW03 (VT/WH) | —             | Ground  |



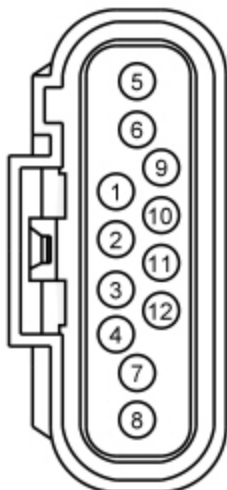
Is the voltage greater than 11 volts?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">A6</a> . |
| No  | REPAIR the circuit.        |

#### A6 CHECK THE WINDSHIELD WIPER MOTOR GROUND CIRCUIT FOR AN OPEN

- Measure the **voltage** between:

| Positive Lead              |               | Negative Lead              |               |
|----------------------------|---------------|----------------------------|---------------|
| Pin                        | Circuit       | Pin                        | Circuit       |
| <a href="#">C125</a> Pin 8 | CRW03 (VT/WH) | <a href="#">C125</a> Pin 6 | GD120 (BK/GN) |



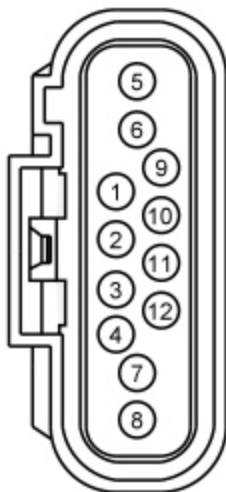
Is the voltage greater than 11 volts?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">A7</a> . |
| No  | REPAIR the circuit.        |

#### A7 CHECK THE FAIL-SAFE HIGH SPEED WIPER GROUND CIRCUIT CONTROL

- Ignition OFF.
- Place the multifunction switch to the high speed setting.
- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead |         |
|----------------------------|---------------|---------------|---------|
| Pin                        | Circuit       | Pin           | Circuit |
| <a href="#">C125</a> Pin 1 | CRW15 (BN/WH) | —             | Ground  |



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Is the resistance less than 3 ohms?

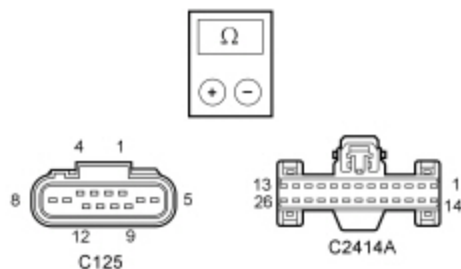
|     |                                                             |
|-----|-------------------------------------------------------------|
| Yes | INSTALL a new windshield wiper motor. REFER to Wiper Motor. |
| No  | GO to <a href="#">A8</a> .                                  |

**A8 CHECK THE FAIL-SAFE HIGH SPEED WIPER GROUND CIRCUIT FOR AN OPEN**

**NOTE:** If steps A1-A7 did not identify the cause of the concern, a double fault is present. After the fail-safe ground is diagnosed and repaired, it is necessary check all operation modes and diagnose any symptom still present.

- Disconnect: SCCM [C2414A](#) .
- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead                 |               |
|----------------------------|---------------|-------------------------------|---------------|
| Pin                        | Circuit       | Pin                           | Circuit       |
| <a href="#">C125</a> Pin 1 | CRW15 (BN/WH) | <a href="#">C2414A</a> Pin 12 | CRW15 (BN/WH) |



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Is the resistance less than 3 ohms?

|     |                                                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | INSTALL a new multifunction switch. REFER to Section 211-05. TEST the system for normal operation. If the concern is still present, GO to <a href="#">A9</a> . |
| No  | REPAIR the circuit.                                                                                                                                            |

## A9 CHECK FOR CORRECT SCCM OPERATION

- Disconnect and inspect all **SCCM** connectors.
- Repair:
  - corrosion (install new connector or terminals – clean module pins)
  - damaged or bent pins – install new terminals/pins
  - pushed-out pins – install new pins as necessary
- Reconnect the **SCCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

### Is the concern still present?

|            |                                                                                                                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | CHECK On-Line Automotive Service Information System (OASIS) for any applicable <b>TSB</b> s . If a <b>TSB</b> exists for this concern, DISCONTINUE this test and FOLLOW <b>TSB</b> instructions. If no <b>TSB</b> s address this concern, INSTALL a new <b>SCCM</b> . REFER to Section 211-05. |
| <b>No</b>  | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.                                                                                                                                 |

## Pinpoint Test B: The Intermittent And/Or Low Speed Windshield Wipers Are Inoperative

Refer to Wiring Diagrams Cell [81](#) , Wipers and Washers for schematic and connector information.

### Normal Operation

The wiper relay (located in the Battery Junction Box (BJB) ) is energized when the ignition is ON. The wiper relay provides voltage to the wiper motor. The multifunction switch is mounted to the Steering Column Control Module (SCCM) . The **SCCM** commands the wiper motor to activate the appropriate wash, low, high and intermittent modes through a Local Interconnect Network (LIN) .

When the multifunction switch is in the HIGH position, the wiper motor receives ground which is hardwired directly through the **SCCM** (bypassing the logic within the **SCCM** and the wiper motor). This fail-safe circuit makes sure that the wipers function in HIGH speed mode in the event of a **LIN** communication failure.

- DTC B1131:02 (Wiper Motor Module: General Signal Failure) — Sets when the **SCCM** detects a general signal failure occurs on the **LIN** .
- DTC B1131:08 (Wiper Motor Module: Bus Signal / Message Failure) — Sets when the **SCCM** loses communication with the windshield wiper motor over the **LIN** .

### This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Multifunction switch
- Windshield wiper motor
- Rain sensor (if equipped)
- **SCCM**

## PINPOINT TEST B : THE INTERMITTENT AND/OR LOW SPEED WINDSHIELD WIPERS ARE INOPERATIVE

**NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.**

**B1 CHECK FOR SCCM DTC B1131:02 OR DTC B1131:08**

- Ignition ON.
- Using a scan tool, perform the SCCM self-test.

**Is DTC B1131:02 or DTC B1131:08 present?**

|            |                            |
|------------|----------------------------|
| <b>Yes</b> | GO to <a href="#">B4</a> . |
| <b>No</b>  | GO to <a href="#">B2</a> . |

**B2 CHECK THE LOW SPEED WINDSHIELD WIPER OPERATION**

- Place the multifunction switch to the low speed setting and observe the windshield wipers. Place the multifunction switch to the off setting.

**Do the windshield wipers operate with the multifunction switch in the low speed setting?**

|            |                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | If not equipped with a rain sensor, <a href="#">GO to Pinpoint Test D</a> .<br>If equipped with a rain sensor, <a href="#">GO to Pinpoint Test E</a> . |
| <b>No</b>  | GO to <a href="#">B3</a> .                                                                                                                             |

**B3 CHECK THE MULTIFUNCTION SWITCH LOW SPEED SETTING INPUT**

- Using a scan tool, monitor the SCCM F\_WIP\_SLO\_SW PID.
- Place the multifunction switch to the low speed setting.

**Does the PID indicate the multifunction switch is in the low speed setting?**

|            |                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | INSTALL a new windshield wiper motor. REFER to Wiper Motor.                                                                                                                                 |
| <b>No</b>  | INSTALL a new multifunction switch. REFER to Section 211-05. TEST the system for normal operation.<br>If the concern is still present, INSTALL a new <u>SCCM</u> . REFER to Section 211-05. |

**B4 CHECK THE HIGH SPEED WIPER OPERATION**

- Place the multifunction switch to the high speed setting and observe the windshield wipers. Place the multifunction switch to the off setting.

**Do the windshield wipers operate with the multifunction switch in the high speed setting?**

|            |                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | If equipped with a rain sensor, GO to <a href="#">B5</a> .<br>If not equipped with a rain sensor, GO to <a href="#">B6</a> . |
| <b>No</b>  | <a href="#">GO to Pinpoint Test A</a> .                                                                                      |

**B5 ISOLATE THE RAIN SENSOR**

- Ignition OFF.
- Disconnect: Rain Sensor [C914](#).
- Ignition ON.
- **NOTE:** *DTC B10AD:02 or DTC B10AD:08 may set and should be ignored.*

Using a scan tool, clear the DTCs and then perform the **SCCM** self-test.

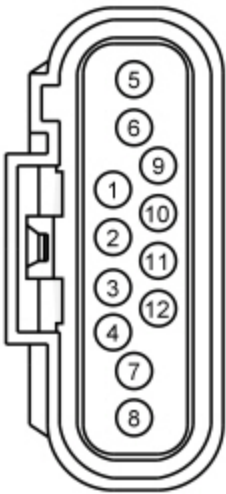
**Is DTC B1131:02 or DTC B1131:08 still present?**

|            |                                                  |
|------------|--------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">B6</a> .                       |
| <b>No</b>  | INSTALL a new rain sensor. REFER to Rain Sensor. |

**B6 CHECK THE WINDSHIELD WIPER MOTOR LOGIC GROUND CIRCUIT FOR AN OPEN**

- Ignition OFF.
- Disconnect: Windshield Wiper Motor [C125](#).
- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead |         |
|----------------------------|---------------|---------------|---------|
| Pin                        | Circuit       | Pin           | Circuit |
| <a href="#">C125</a> Pin 3 | GD120 (BK/GN) | —             | Ground  |



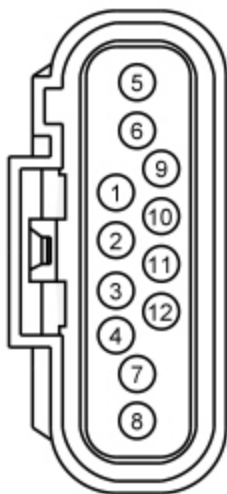
**Is the resistance less than 3 ohms?**

|            |                            |
|------------|----------------------------|
| <b>Yes</b> | GO to <a href="#">B7</a> . |
| <b>No</b>  | REPAIR the circuit.        |

## B7 CHECK THE LIN CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: SCCM [C2414A](#) .
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead              |               | Negative Lead |         |
|----------------------------|---------------|---------------|---------|
| Pin                        | Circuit       | Pin           | Circuit |
| <a href="#">C125</a> Pin 4 | VDN05 (YE/GY) | —             | Ground  |



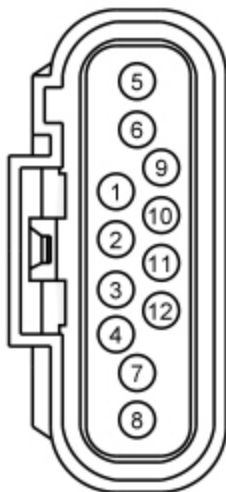
Is any voltage present?

|            |                            |
|------------|----------------------------|
| <b>Yes</b> | REPAIR the circuit.        |
| <b>No</b>  | GO to <a href="#">B8</a> . |

## B8 CHECK THE LIN CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead |         |
|----------------------------|---------------|---------------|---------|
| Pin                        | Circuit       | Pin           | Circuit |
| <a href="#">C125</a> Pin 4 | VDN05 (YE/GY) | —             | Ground  |



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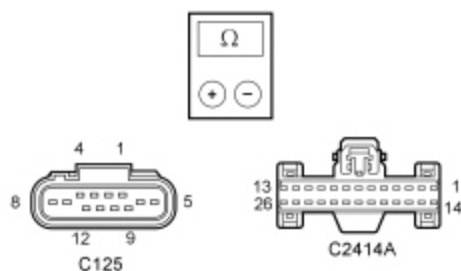
Is the resistance greater than 10,000 ohms?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">B9</a> . |
| No  | REPAIR the circuit.        |

## B9 CHECK THE LIN CIRCUIT FOR AN OPEN

- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead                 |               |
|----------------------------|---------------|-------------------------------|---------------|
| Pin                        | Circuit       | Pin                           | Circuit       |
| <a href="#">C125</a> Pin 4 | VDN05 (YE/GY) | <a href="#">C2414A</a> Pin 20 | VDN05 (YE/GY) |



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Is the resistance less than 3 ohms?

|     |                                                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | INSTALL a new windshield wiper motor. REFER to Wiper Motor. TEST the system for normal operation. If the concern is still present, GO to <a href="#">B10</a> . |
| No  | REPAIR the circuit.                                                                                                                                            |

## B10 CHECK FOR CORRECT SCCM OPERATION

- Disconnect and inspect all ~~SCCM~~ connectors.



- Repair:
  - corrosion (install new connector or terminals – clean module pins)
  - damaged or bent pins – install new terminals/pins
  - pushed-out pins – install new pins as necessary
- Reconnect the **SCCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

#### Is the concern still present?

|            |                                                                                                                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | CHECK On-Line Automotive Service Information System (OASIS) for any applicable <b>TSB</b> s . If a <b>TSB</b> exists for this concern, DISCONTINUE this test and FOLLOW <b>TSB</b> instructions. If no <b>TSB</b> s address this concern, INSTALL a new <b>SCCM</b> . REFER to Section 211-05. |
| <b>No</b>  | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.                                                                                                                                 |

#### Pinpoint Test C: The Windshield Wipers Stay On Continuously

Refer to Wiring Diagrams Cell [81](#) , Wipers and Washers for schematic and connector information.

#### Normal Operation

The wiper relay (located in the Battery Junction Box (BJB) ) is energized when the ignition is ON. The wiper relay provides voltage to the wiper motor. The multifunction switch is mounted to the Steering Column Control Module (SCCM) . The **SCCM** commands the wiper motor to activate the appropriate wash, low, high and intermittent modes through a Local Interconnect Network (LIN) .

When the multifunction switch is in the HIGH position, the wiper motor receives ground which is hardwired directly through the **SCCM** (bypassing the logic within the **SCCM** and the wiper motor). This fail-safe circuit makes sure that the wipers function in HIGH speed mode in the event of a **LIN** communication failure.

- DTC B1008:09 (Wiper Mode Switch: Component Failure) — Sets when the **SCCM** detects a fault from the wiper/washer switch input.

#### This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Wiper relay
- Multifunction switch
- Wiper motor
- **SCCM**

#### PINPOINT TEST C : THE WINDSHIELD WIPERS STAY ON CONTINUOUSLY

|                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOTICE:</b> Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector. |
| <b>C1 CHECK THE WINDSHIELD WIPER OPERATION WITH THE IGNITION OFF</b>                                                                                                    |

- Ignition OFF.
- Observe the windshield wiper operation.

Are the windshield wipers operating?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">C2</a> . |
| No  | GO to <a href="#">C3</a> . |

C2 CHECK THE WIPER RELAY OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: Wiper Relay .
- Observe the windshield wiper operation.

Do the windshield wipers continue to operate?

|     |                            |
|-----|----------------------------|
| Yes | REPAIR the circuit.        |
| No  | INSTALL a new wiper relay. |

C3 CHECK THE WINDSHIELD WIPER OPERATION WITH THE SCCM DISCONNECTED

- Disconnect: ~~SCCM~~ [C2414A](#) .
- Ignition ON.
- Observe the windshield wiper operation.

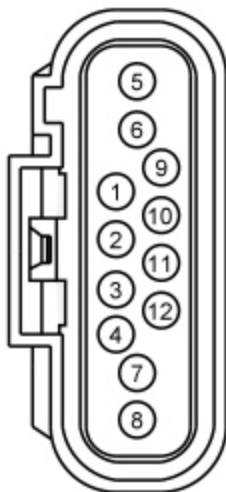
Do the windshield wipers continue to operate?

|     |                                                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | GO to <a href="#">C4</a> .                                                                                                                                     |
| No  | INSTALL a new multifunction switch. REFER to Section 211-05. TEST the system for normal operation. If the concern is still present, GO to <a href="#">C5</a> . |

C4 CHECK THE HIGH SPEED FAIL-SAFE CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: Windshield Wiper Motor [C125](#).
- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead |         |
|----------------------------|---------------|---------------|---------|
| Pin                        | Circuit       | Pin           | Circuit |
| <a href="#">C125</a> Pin 1 | CRW15 (BN/WH) | —             | Ground  |



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**Is the resistance greater than 10,000 ohms?**

|            |                                                             |
|------------|-------------------------------------------------------------|
| <b>Yes</b> | INSTALL a new windshield wiper motor. REFER to Wiper Motor. |
| <b>No</b>  | REPAIR the circuit.                                         |

## C5 CHECK FOR CORRECT SCCM OPERATION

- Disconnect and inspect all **SCCM** connectors.
- Repair:
  - corrosion (install new connector or terminals – clean module pins)
  - damaged or bent pins – install new terminals/pins
  - pushed-out pins – install new pins as necessary
- Reconnect the **SCCM** connectors. Make sure they seat and latch correctly.
- Operate the system and determine if the concern is still present.

**Is the concern still present?**

|            |                                                                                                                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | CHECK On-Line Automotive Service Information System (OASIS) for any applicable <b>TSBs</b> . If a <b>TSB</b> exists for this concern, DISCONTINUE this test and FOLLOW <b>TSB</b> instructions. If no <b>TSBs</b> address this concern, INSTALL a new <b>SCCM</b> . REFER to Section 211-05. |
| <b>No</b>  | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.                                                                                                                               |

## Pinpoint Test D: The Windshield Wipers Do Not Operate Correctly In The Intermittent Mode

### Normal Operation

The wiper relay (located in the Battery Junction Box (BJB) ) is energized when the ignition is ON. The wiper relay provides voltage to the wiper motor. The multifunction switch is mounted to the Steering Column Control Module (SCCM) . The **SCCM** commands the wiper motor to activate the appropriate wash, low, high and intermittent modes through a Local Interconnect Network (LIN) .

**This pinpoint test is intended to diagnose the following:**

- Multifunction switch
- Windshield wiper motor
- SCCM

## PINPOINT TEST D : THE WINDSHIELD WIPERS DO NOT OPERATE CORRECTLY IN THE INTERMITTENT MODE

### D1 VERIFY IF THE VEHICLE IS EQUIPPED WITH A RAIN SENSOR

- Inspect the vehicle for a rain sensor.

**Is the vehicle equipped with a rain sensor?**

|            |                            |
|------------|----------------------------|
| <b>Yes</b> | GO to <a href="#">D2</a> . |
| <b>No</b>  | GO to <a href="#">D3</a> . |

### D2 VERIFY THE RAIN SENSITIVE WIPER FEATURE IS DISABLED IN THE VEHICLE MESSAGE CENTER

- Ignition ON.
- Access the vehicle settings in the message center using the steering wheel controls and check the rain sensitive feature setting.

**Is the rain sensitive feature enabled?**

|            |                                         |
|------------|-----------------------------------------|
| <b>Yes</b> | <a href="#">GO to Pinpoint Test E</a> . |
| <b>No</b>  | GO to <a href="#">D3</a> .              |

### D3 CHECK THE LOW SPEED WIPER OPERATION

- Ignition ON.
- Place the multifunction switch to the low speed setting and observe the windshield wipers. Place the multifunction switch to the off setting.

**Do the low speed wipers operate when the low speed setting is selected?**

|            |                                         |
|------------|-----------------------------------------|
| <b>Yes</b> | GO to <a href="#">D4</a> .              |
| <b>No</b>  | <a href="#">GO to Pinpoint Test B</a> . |

### D4 CHECK THE INPUT FROM THE MULTIFUNCTION SWITCH

- Using a scan tool, monitor the SCCM multifunction switch PIDs:
  - F\_WIP\_INT\_D1
  - F\_WIP\_INT\_D2
  - F\_WIP\_INT\_D3
  - F\_WIP\_INT\_D4
  - F\_WIP\_INT\_D5
- Cycle the multifunction switch from off through each of the intermittent settings.

**Do the PIDs correspond correctly with each setting selection?**

|            |                                                                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | INSTALL a new windshield wiper motor. REFER to Wiper Motor.                                                                                                                      |
| <b>No</b>  | INSTALL a new multifunction switch. REFER to Section 211-05. TEST the system for normal operation. If the concern is still present, INSTALL a new SCCM. REFER to Section 211-05. |

**Pinpoint Test E: The Rain Sensitive Wiper Feature Is Inoperative**

Refer to Wiring Diagrams Cell [81](#) , Wipers and Washers for schematic and connector information.

**Normal Operation**

The rain sensitive wiper feature replaces the intermittent wiper feature. The rain sensitive feature is active when the multifunction switch is in any of the 5 auto/intermittent sensitivity settings. The setting closest to OFF is the least sensitive to moisture, the highest setting being the most sensitive to moisture.

The rain sensor sends data through the LIN to the windshield wiper motor. When the rain sensitive wiper feature is enabled, based on the data received from the rain sensor and the auto/intermittent setting on the multifunction switch, the system automatically activates and adjusts the wiper speed and frequency when moisture is detected on the windshield.

The rain sensor optically senses water on the windshield. An infrared beam lights a section of the windshield. The intensity of the reflected beam is measured and compared to the intensity of the transmitted beam. If there is water on the surface of the windshield, the beam distorts, reducing the intensity of the beam. If the beam is not reflected completely, it is interpreted as rain on the windshield and the windshield wipers are activated.

A gel-type lens is used as the interface between the rain sensor and the windshield. If the lens is missing or damaged, the rain sensing wipers can be inoperative. The rain sensor gel-type lens is part of the rain sensor and cannot be replaced separately on a rain sensor of this type.

The rain sensitive wiper feature can be enabled/disabled through the message center.

When the rain sensitive wipers are disabled from the message center, or the windshield wiper motor detects a fault with the rain sensor input, the windshield wiper motor changes to a default intermittent mode. The timing of the wipers correspond to a standard intermittent setting based on the auto/intermittent sensitivity setting on the multifunction switch.

- B10AD:02 (Rain Sensor: General Signal Failure) — This DTC sets when the SCCM determines a general signal failure has occurred on the LIN .
- B10AD:08 (Rain Sensor: Bus Signal / Message Failure) — This DTC sets when the SCCM determines a communication error has occurred on the LIN . If the rain sensitive wipers are inoperative only, this DTC may be caused by an open in the rain sensor power or ground circuit or an open in the LIN circuit.

**This pinpoint test is intended to diagnose the following:**

- Fuse
- Wiring, terminals or connectors
- Rain sensitive feature not enabled
- Foreign material on windshield, such as dirt, dust or bugs
- Wiper blades
- Windshield glass
- Multifunction switch
- Windshield wiper motor
- Rain sensor
- SCCM

## PINPOINT TEST E : THE RAIN SENSITIVE WIPER FEATURE IS INOPERATIVE

### E1 CHECK THE LOW SPEED WIPER OPERATION

**NOTICE:** Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

- Ignition ON.
- Place the multifunction switch to the low speed setting and observe the windshield wipers. Place the multifunction switch to the off setting.

**Do the low speed wipers operate when the low speed setting is selected?**

|            |                                         |
|------------|-----------------------------------------|
| <b>Yes</b> | GO to <a href="#">E2</a> .              |
| <b>No</b>  | <a href="#">GO to Pinpoint Test B</a> . |

### E2 CHECK FOR DTC B1131:02 OR DTC B1131:08

- Ignition ON.
- Using a scan tool, perform the SCCM self-test.

**Is DTC B1131:02 or DTC B1131:08 present?**

|            |                                         |
|------------|-----------------------------------------|
| <b>Yes</b> | <a href="#">GO to Pinpoint Test B</a> . |
| <b>No</b>  | GO to <a href="#">E3</a> .              |

### E3 CHECK FOR DTC B10AD:02 OR DTC B10AD:08

- Review the results from the SCCM self-test.

**Is DTC B10AD:02 or DTC B10AD:08 present?**

|            |                             |
|------------|-----------------------------|
| <b>Yes</b> | GO to <a href="#">E12</a> . |
| <b>No</b>  | GO to <a href="#">E4</a> .  |

### E4 VERIFY THE RAIN SENSITIVE WIPER FEATURE IS ENABLED IN THE VEHICLE MESSAGE CENTER

- Access the vehicle settings in the message center using the steering wheel controls. Make sure the rain sensitive feature is enabled.

**Is the rain sensitive feature enabled?**

|            |                                                                                                          |
|------------|----------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">E5</a> .                                                                               |
| <b>No</b>  | ENABLE the rain sensitive feature. INFORM the customer how to enable/disable the rain sensitive feature. |

## E5 CHECK THE INPUT FROM THE MULTIFUNCTION SWITCH

- Using a scan tool, monitor the SCCM multifunction switch PIDs:
  - F\_WIP\_INT\_D1
  - F\_WIP\_INT\_D2
  - F\_WIP\_INT\_D3
  - F\_WIP\_INT\_D4
  - F\_WIP\_INT\_D5
- Cycle the multifunction switch from off through each of the auto/intermittent sensitivity settings.

**Do the PIDs correspond correctly with each setting selection?**

|            |                                                                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">E6</a> .                                                                                                                                                               |
| <b>No</b>  | INSTALL a new multifunction switch. REFER to Section 211-05. TEST the system for normal operation. If the concern is still present, INSTALL a new <u>SCCM</u> . REFER to Section 211-05. |

## E6 CHECK THE WINDSHIELD FOR DAMAGE

- Ignition OFF.
- Inspect the windshield glass for any cracks, pits or other visible damage around the area of the rain sensor.

**Is any windshield damage present near the area of the rain sensor?**

|            |                                                    |
|------------|----------------------------------------------------|
| <b>Yes</b> | INSTALL a new windshield. REFER to Section 501-11. |
| <b>No</b>  | GO to <a href="#">E7</a> .                         |

## E7 CHECK THE WINDSHIELD GLASS FOR STREAKING

- Ignition ON.
- Turn the multifunction switch to the low speed wiper setting and while applying water to the windshield surface. Observe the wipe quality, especially around the area or the rain sensor.

**Is any streaking present on the windshield around the area of the rain sensor?**

|            |                             |
|------------|-----------------------------|
| <b>Yes</b> | GO to <a href="#">E8</a> .  |
| <b>No</b>  | GO to <a href="#">E10</a> . |

## E8 CHECK THE WINDSHIELD WIPER BLADES FOR STREAKING

- Ignition OFF.
- Clean the windshield glass with glass cleaner.
- Clean the wiper blades with isopropyl alcohol swabs. Wipe the entire length of the wiper blades.
- Ignition ON.

- Place the multifunction switch to the low speed wiper setting and while applying water to the windshield surface. Observe the wipe quality, especially around the area or the rain sensor.

**Was there noticeable improvement in the wipe quality after cleaning the glass and the wiper blades?**

|            |                                                      |
|------------|------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">E9</a> .                           |
| <b>No</b>  | INSTALL new wiper blades. GO to <a href="#">E9</a> . |

## **E9 RECHECK THE RAIN SENSITIVE WIPERS OPERATION**

- Place the multifunction switch to the auto/intermittent 5 setting (most sensitive to moisture) while applying water on the windshield above the rain sensor.

**Do the wipers operate when water is applied to the windshield surface?**

|            |                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | The system is operating correctly at this time. The concern may have been caused by poor wipe quality on the windshield around the rain sensor. |
| <b>No</b>  | GO to <a href="#">E10</a> .                                                                                                                     |

## **E10 CHECK FOR THE PRESENCE OF THE LENS GEL**

- Ignition OFF.
- Lightly press the rain sensor against the windshield and release the rain sensor clips.

**Does the rain sensor remain adhered to the windshield?**

|            |                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------|
| <b>Yes</b> | ATTACH the retaining clips. GO to <a href="#">E11</a> .                                          |
| <b>No</b>  | The sensor gel-type lens is missing or damaged. INSTALL a new rain sensor. REFER to Rain Sensor. |

## **E11 INSPECT THE RAIN SENSOR MOUNTING AREA**

**NOTE:** *The rain sensor gel-type lens can be damaged or contaminated during rain sensor removal, rain sensor bracket separation or incorrect installation.*

- From the outside of the vehicle, inspect the area of the windshield glass where the rain sensor is mounted. The rain sensor contact area should appear black, with no bubbles, air pockets or contaminants between the windshield glass and the rain sensor.

**Does the rain sensor contact area appear black, with no bubbles, air pockets or contaminants between the windshield glass and the rain sensor?**

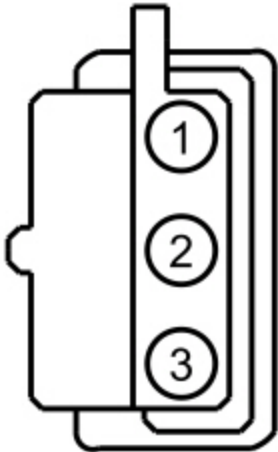
|            |                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">E12</a> .                                                                                                                                               |
| <b>No</b>  | REMOVE and INSTALL the rain sensor correctly. REFER to Rain Sensor. TEST the system for normal operation.<br>If the concern is still present, GO to <a href="#">E15</a> . |



E12 CHECK THE RAIN SENSOR VOLTAGE SUPPLY CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: Rain Sensor [C914](#).
- Ignition ON.
- Measure the **voltage** between:

| Positive Lead              |            | Negative Lead |         |
|----------------------------|------------|---------------|---------|
| Pin                        | Circuit    | Pin           | Circuit |
| <a href="#">C914</a> Pin 1 | SBB79 (YE) | —             | Ground  |



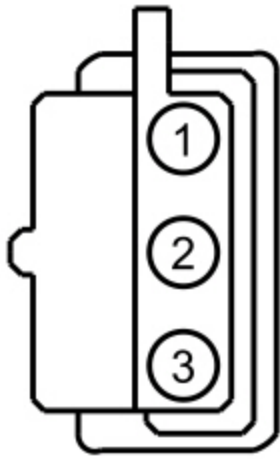
Is the voltage greater than 11 volts?

|     |                                                                                                                                                                           |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | GO to <a href="#">E13</a> .                                                                                                                                               |
| No  | VERIFY <del>BJB</del> fuse 79 (5A) is OK. If OK, REPAIR the circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

E13 CHECK THE RAIN SENSOR GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the **resistance** between:

| Positive Lead              |            | Negative Lead |         |
|----------------------------|------------|---------------|---------|
| Pin                        | Circuit    | Pin           | Circuit |
| <a href="#">C914</a> Pin 3 | GD133 (BK) | —             | Ground  |



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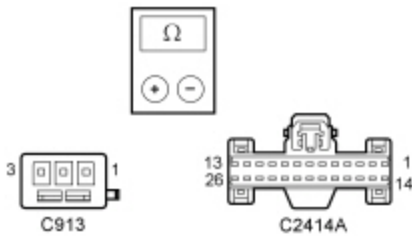
Is the resistance less than 3 ohms?

|     |                             |
|-----|-----------------------------|
| Yes | GO to <a href="#">E14</a> . |
| No  | REPAIR the circuit.         |

#### E14 CHECK THE LIN CIRCUIT FOR AN OPEN

- Disconnect: SCCM [C2414B](#) .
- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead                 |               |
|----------------------------|---------------|-------------------------------|---------------|
| Pin                        | Circuit       | Pin                           | Circuit       |
| <a href="#">C914</a> Pin 2 | VDN05 (YE/GY) | <a href="#">C2414A</a> Pin 20 | VDN05 (YE/GY) |



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Is the resistance less than 3 ohms?

|     |                             |
|-----|-----------------------------|
| Yes | GO to <a href="#">E15</a> . |
| No  | REPAIR the circuit.         |

#### E15 CHECK FOR CORRECT RAIN SENSOR OPERATION

- Inspect the rain sensor connector.

- Repair:
  - corrosion (install new connector or terminals – clean module pins)
  - damaged or bent pins – install new terminals/pins
  - pushed-out pins – install new pins as necessary
- Reconnect the rain sensor connector. Make sure it seats and latches correctly.
- Operate the system and determine if the concern is still present.

#### Is the concern still present?

|            |                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Yes</b> | INSTALL a new rain sensor. REFER to Rain Sensor. TEST the system for normal operation.<br>If the concern is still present, INSTALL a new windshield wiper motor. REFER to Wiper Motor. |
| <b>No</b>  | The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.                         |

### Pinpoint Test F: The Windshield Wipers Do Not Park At The Correct Position

#### Normal Operation

When the multifunction switch is placed in the OFF position, the motor continues to operate until the motor returns to the PARK position and the internal Hall-effect sensor senses the motor magnet. The output to the integral run/park relay deactivates the relay and disconnects the voltage to the motor.

#### This pinpoint test is intended to diagnose the following:

- Wiper mounting arm and pivot shaft
- Wiper motor

#### PINPOINT TEST F : THE WINDSHIELD WIPERS DO NOT PARK AT THE CORRECT POSITION

##### F1 CHECK THE WIPER MOUNTING ARE AND PIVOT SHAFT FOR DAMAGE

- Ignition OFF.
- Inspect the wiper mounting arm and pivot shaft for damage.

#### Is the wiper mounting arm and pivot shaft OK?

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">F2</a> .                                                                 |
| <b>No</b>  | INSTALL a wiper mounting arm and pivot shaft. REFER to Wiper Mounting Arm and Pivot Shaft. |

##### F2 CHECK THE WINDSHIELD WIPER PARK RELAY OPERATION

- Ignition ON.
- Place the multifunction switch to the low speed setting. When the wiper blades are in the middle of a wipe, place the multifunction switch off.

#### Do the windshield wipers immediately stop when the multifunction switch is turned off?

|     |                                                                       |
|-----|-----------------------------------------------------------------------|
| Yes | INSTALL a new windshield wiper motor. REFER to Wiper Motor.           |
| No  | ADJUST the pivot arms. REFER to Wiper Blade and Pivot Arm Adjustment. |

**Pinpoint Test G: The Washer Pump Is Inoperative Or Always On - Vehicles Not Equipped With Front Video Camera**

Refer to Wiring Diagrams Cell [81](#) , Wipers and Washers for schematic and connector information.

**Normal Operation**

When the multifunction switch is placed in the WASH position, the Steering Column Control Module (SCCM) sends the request to the wiper motor over the Local Interconnect Network (LIN) . The wiper motor then supplies power to the washer pump, which directs washer fluid to the windshield.

**This pinpoint test is intended to diagnose the following:**

- Wiring, terminals or connectors
- Washer pump
- Windshield wiper motor
- SCCM

**PINPOINT TEST G : THE WASHER PUMP IS INOPERATIVE OR ALWAYS ON - VEHICLES NOT EQUIPPED WITH FRONT VIDEO CAMERA**

**NOTICE:** Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

**G1 CHECK THE MULTIFUNCTION SWITCH INPUT**

- Ignition ON.
- Using a scan tool, monitor the SCCM F\_WASH\_SW PID.
- Press and release the washer switch on the multifunction switch.

**Does the PID indicate correctly when the switch pressed and released?**

|     |                                                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | If the washer pump is always on, GO to <a href="#">G2</a> .<br>If the washer pump is inoperative, GO to <a href="#">G3</a> .                   |
| No  | INSTALL a new multifunction switch. REFER to Section 211-05.<br>If the concern is still present, INSTALL a new SCCM . REFER to Section 211-05. |

**G2 CHECK THE WASHER PUMP SUPPLY CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Windshield Wiper Motor [C125](#).
- Ignition ON.

**Does the washer pump continue to operate?**

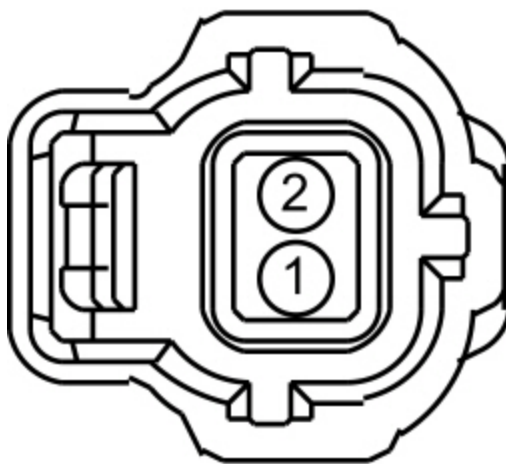
|     |                     |
|-----|---------------------|
| Yes | REPAIR the circuit. |
|-----|---------------------|

|    |                                                             |
|----|-------------------------------------------------------------|
| No | INSTALL a new windshield wiper motor. REFER to Wiper Motor. |
|----|-------------------------------------------------------------|

### G3 CHECK THE WASHER PUMP CONTROL CIRCUIT INTEGRITY

- Ignition OFF.
- Disconnect: Washer Pump [C137](#).
- Connect a **test lamp** between:

| Pin                        | Circuit       | Pin | Circuit |
|----------------------------|---------------|-----|---------|
| <a href="#">C137</a> Pin 1 | CRW14 (BU/WH) | —   | Ground  |



- Ignition ON.
- Observe the test lamp while placing the multifunction switch into the windshield WASH position.

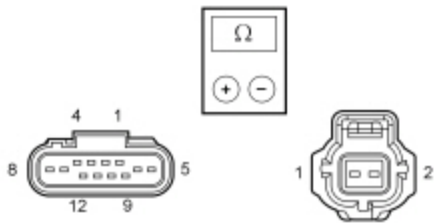
Does the test lamp illuminate?

|     |                            |
|-----|----------------------------|
| Yes | GO to <a href="#">G5</a> . |
| No  | GO to <a href="#">G4</a> . |

### G4 CHECK THE WASHER PUMP CONTROL CIRCUIT FOR AN OPEN

- Measure the **resistance** between:

| Positive Lead              |               | Negative Lead              |               |
|----------------------------|---------------|----------------------------|---------------|
| Pin                        | Circuit       | Pin                        | Circuit       |
| <a href="#">C125</a> Pin 7 | CRW14 (BU/WH) | <a href="#">C137</a> Pin 1 | CRW14 (BU/WH) |



- N0138461

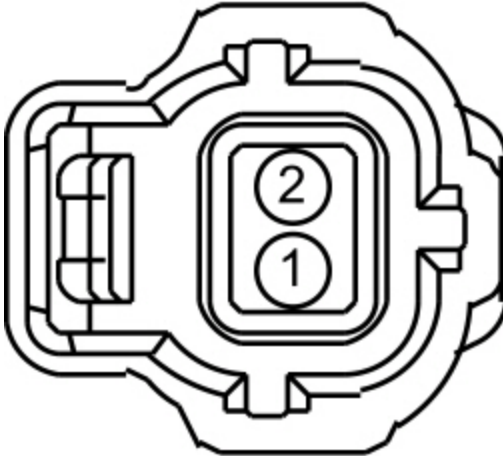
Is the resistance less than 3 ohms?

|     |                                                             |
|-----|-------------------------------------------------------------|
| Yes | INSTALL a new windshield wiper motor. REFER to Wiper Motor. |
| No  | REPAIR the circuit.                                         |

G5 CHECK THE WASHER PUMP GROUND CIRCUIT FOR AN OPEN

- Connect a **test lamp** between:

| Pin | Circuit               | Pin                        | Circuit       |
|-----|-----------------------|----------------------------|---------------|
| —   | Positive battery post | <a href="#">C137</a> Pin 2 | GD120 (BK/GN) |



- 

Does the test lamp illuminate?

|     |                                                  |
|-----|--------------------------------------------------|
| Yes | INSTALL a new washer pump. REFER to Washer Pump. |
| No  | REPAIR the circuit.                              |

Pinpoint Test H: The Washer Pump is Inoperative Or Always On — Vehicles Equipped With Front Video Camera

Refer to Wiring Diagrams Cell [81](#) , Wipers and Washers for schematic and connector information.

Normal Operation

When the multifunction switch is placed in the WASH position, the Steering Column Control Module (SCCM) sends the request to the windshield wiper motor through the Local Interconnect Network (LIN) . The wiper motor then supplies power to the washer pump, which directs washer fluid to the windshield. When the front camera washer control is activated, the Steering Column Control Module (SCCM) sends the request to the windshield wiper motor through the Local Interconnect Network (LIN) . The windshield wiper motor controls the direction of the washer pump to direct washer fluid to the front video camera.

When a windshield wash is requested, the windshield wiper motor supplies voltage to the washer pump. Ground is provided to the washer pump through the normally closed contact in the front camera washer relay (located in the auxiliary BJB ).

When a front camera wash is requested, the windshield wiper motor activates the front camera washer relay. When the front camera washer relay is energized, it provides voltage to the washer pump. Ground is provided to the washer pump through the windshield wiper motor.

This pinpoint test is intended to diagnose the following:

- Fuse
- Front camera washer relay
- SCCM
- Washer pump
- Wiring, terminals or connectors
- Multifunction switch
- Windshield wiper motor

PINPOINT TEST H : THE WASHER PUMP IS INOPERATIVE OR ALWAYS ON — VEHICLES EQUIPPED WITH FRONT VIDEO CAMERA

**NOTICE:** Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

H1 CHECK THE MULTIFUNCTION SWITCH (F\_WASH\_SW) AND (F\_CAMRA\_WASHSW) PIDS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — SCCM .
- While monitoring the SCCM F\_WASH\_SW PID, place the multifunction switch into the windshield wiper WASH position.
- While monitoring the SCCM F\_CAMRA\_WASHSW PID, rotate the front camera washer control to the WASH position.

Do each of the PIDs change to the ON state when the multifunction switch and front camera washer control are in the applicable WASH positions?

|     |                                                                                                                                                                                                                                    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | If a washer function is inoperative, GO to <a href="#">H2</a> .<br>If the windshield washer function is always on, GO to <a href="#">H12</a> .<br>If the front camera washer function is always on, GO to <a href="#">H13</a> .    |
| No  | INSTALL a new multifunction switch. REFER to section Section 211-05. TEST the system for normal operation.<br>If the concern is still present, INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation. |

H2 CHECK THE WINDSHIELD WASH FUNCTION

- Ignition ON.

- Attempt to operate the windshield wash function.

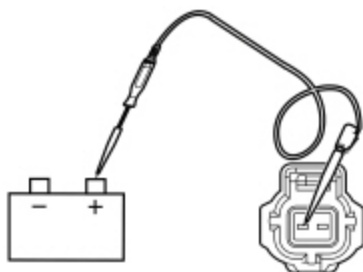
**Does the washer pump operate in the windshield WASH position?**

|            |                            |
|------------|----------------------------|
| <b>Yes</b> | GO to <a href="#">H8</a> . |
| <b>No</b>  | GO to <a href="#">H3</a> . |

### H3 CHECK THE WASHER PUMP CONTROL CIRCUIT FOR AN OPEN

**NOTICE:** The following step uses a test lamp to simulate normal circuit loads. Use only the test lamp recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test lamp probe connection to the vehicle. Do not use the test lamp probe directly on any connector.

- Ignition OFF.
- Disconnect: Washer Pump Motor [C137](#).
- Connect a 12-volt incandescent test lamp between the positive battery post and the washer pump motor [C137](#) Pin 1, circuit CRW24 (BU/WH), harness side.



N0132328

**Does the test lamp illuminate?**

|            |                            |
|------------|----------------------------|
| <b>Yes</b> | GO to <a href="#">H6</a> . |
| <b>No</b>  | GO to <a href="#">H4</a> . |

### H4 CHECK THE FRONT CAMERA WASHER RELAY (WINDSHIELD WASH)

- Disconnect: Front Camera Washer Relay .
- Carry out the front camera washer relay component test.  
Refer to Wiring Diagrams Cell [149](#) for component testing.

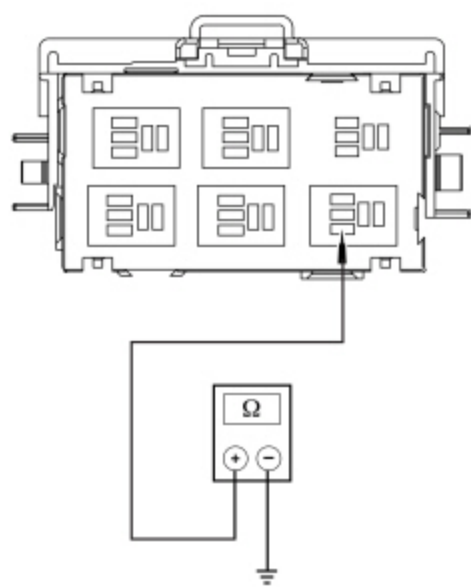
**Did the front camera washer relay pass the component test?**

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">H5</a> .                                                     |
| <b>No</b>  | INSTALL a new front camera washer relay. TEST the system for normal operation. |

### H5 CHECK THE FRONT CAMERA WASHER RELAY GROUND CIRCUIT FOR AN OPEN



- Measure the resistance between the camera washer relay socket pin 4, circuit GD120 (BK/GN) and ground.



• N0132330

Is the resistance less than 3 ohms?

|     |                                                                                 |
|-----|---------------------------------------------------------------------------------|
| Yes | REPAIR circuit CRW24 (BU/WH) for an open. TEST the system for normal operation. |
| No  | REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation. |

**H6 CHECK THE WASHER PUMP CONTROL CIRCUIT FOR AN OPEN**

**NOTICE:** The following step uses a test lamp to simulate normal circuit loads. Use only the test lamp recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test lamp probe connection to the vehicle. Do not use the test lamp probe directly on any connector.

- Ignition ON.
- Connect a 12-volt incandescent test lamp between the washer pump motor [C137](#) Pin 1, circuit CRW24 (BU/WH), harness side and the washer pump motor [C137](#) Pin 2, circuit CRW14 (GY), harness side.



• N0119238

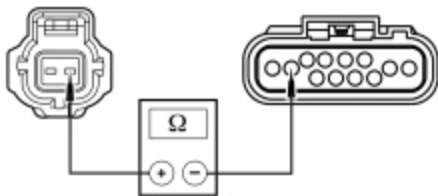
- Observe the test lamp while placing the multifunction switch into the windshield WASH position.

Does the test lamp illuminate when WASH is selected?

|     |                                                                                                        |
|-----|--------------------------------------------------------------------------------------------------------|
| Yes | INSTALL a new washer pump. REFER to Washer Pump in this section. TEST the system for normal operation. |
| No  | GO to <a href="#">H7</a> .                                                                             |

**H7 CHECK THE WASHER PUMP CONTROL CIRCUIT FOR AN OPEN**

- Ignition OFF.
- Disconnect: Windshield Wiper Motor [C125](#).
- Measure the resistance between the washer pump [C137](#) Pin 2, circuit CRW14 (GY), harness side and the windshield wiper motor [C125](#) Pin 7, circuit CRW14 (GY), harness side.



• N0121923

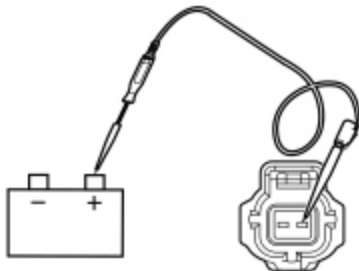
**Is the resistance less than 3 ohms?**

|     |                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------|
| Yes | INSTALL a new windshield wiper motor. REFER to Wiper Motor in this section. TEST the system for normal operation. |
| No  | REPAIR the circuit. TEST the system for normal operation.                                                         |

**H8 CHECK THE WINDSHIELD WIPER MOTOR FOR AN INTERNAL OPEN**

**NOTICE:** The following step uses a test lamp to simulate normal circuit loads. Use only the test lamp recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test lamp probe connection to the vehicle. Do not use the test lamp probe directly on any connector.

- Ignition OFF.
- Disconnect: Washer Pump Motor [C137](#).
- Connect a 12-volt incandescent test lamp between the positive battery post and the washer pump motor [C137](#) Pin 2, circuit CRW14 (GY), harness side.



• N0132327

Does the test lamp illuminate?

|     |                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------|
| Yes | GO to <a href="#">H9</a> .                                                                                        |
| No  | INSTALL a new windshield wiper motor. REFER to Wiper Motor in this section. TEST the system for normal operation. |

**H9 CHECK THE FRONT CAMERA WASHER RELAY (INOPERATIVE)**

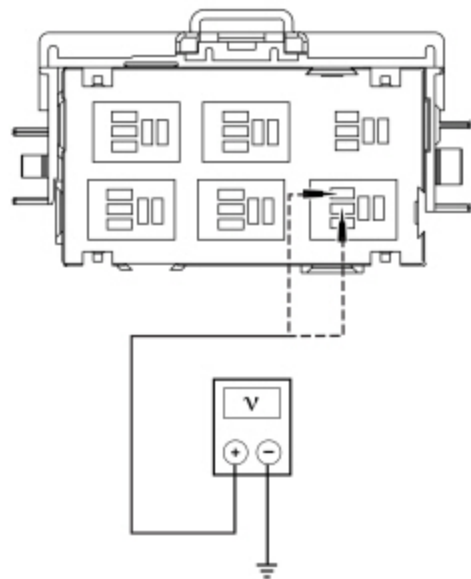
- Disconnect: Front Camera Washer Relay .
- Carry out the front camera washer relay component test.  
Refer to Wiring Diagrams Cell [149](#) for component testing.

Did the front camera washer relay pass the component test?

|     |                                                                                |
|-----|--------------------------------------------------------------------------------|
| Yes | GO to <a href="#">H10</a> .                                                    |
| No  | INSTALL a new front camera washer relay. TEST the system for normal operation. |

**H10 CHECK FOR VOLTAGE TO THE FRONT CAMERA WASHER RELAY**

- Ignition ON.
- Measure the voltage between the front camera washer relay socket pin 1, circuit CBB54 (VT/OG) and ground; and between the front camera washer relay socket pin 5, circuit SBB41 (RD) and ground.



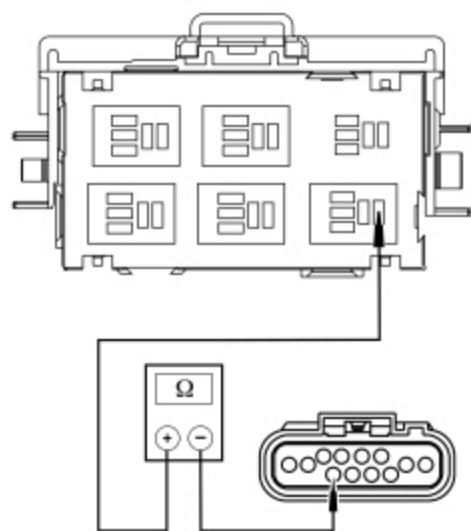
N0132333

Are the voltages greater than 11?

|     |                                                                                                                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes | GO to <a href="#">H11</a> .                                                                                                                                                                                                     |
| No  | VERIFY BJB fuses 41 (15A) and 54 (5A) are OK. If OK, REPAIR the circuit in question. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. |

**H11 CHECK THE FRONT CAMERA WASHER RELAY CONTROL CIRCUIT FOR AN OPEN**

- Ignition OFF.
- Disconnect: Windshield Wiper Motor [C125](#).
- Measure the resistance between the front camera washer relay socket pin 3, circuit CRW34 (BU/BN) and the windshield wiper motor [C125](#) Pin 12, circuit CRW34 (BU/BN), harness side.



• N0133870

**Is the resistance less than 3 ohms?**

|     |                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------|
| Yes | INSTALL a new windshield wiper motor. REFER to Wiper Motor in this section. TEST the system for normal operation. |
| No  | REPAIR the circuit. TEST the system for normal operation.                                                         |

**H12 CHECK THE WASHER PUMP SUPPLY CIRCUIT FROM THE WIPER MOTOR FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Windshield Wiper Motor [C125](#).
- Ignition ON.

**Does the washer pump continue to operate?**

|     |                                                                                         |
|-----|-----------------------------------------------------------------------------------------|
| Yes | REPAIR circuit CRW14 (GY) for a short to voltage. TEST the system for normal operation. |
| No  | INSTALL a new windshield wiper motor. REFER to Wiper Motor.                             |

**H13 CHECK THE FRONT CAMERA WASHER RELAY OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE**

- Ignition OFF.
- Disconnect: Front Camera Washer Relay .
- Ignition ON.

**Does the washer pump continue to operate?**

|            |                                                                                            |
|------------|--------------------------------------------------------------------------------------------|
| <b>Yes</b> | REPAIR circuit CRW24 (BU/WH) for a short to voltage. TEST the system for normal operation. |
| <b>No</b>  | GO to <a href="#">H14</a> .                                                                |

**H14 CHECK THE FRONT CAMERA WASHER RELAY (ALWAYS ON)**

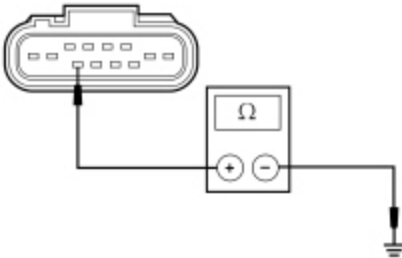
- Ignition OFF.
- Carry out the front camera washer relay component test.  
Refer to Wiring Diagrams Cell [149](#) for component testing.

**Did the front camera washer relay pass the component test?**

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| <b>Yes</b> | GO to <a href="#">H15</a> .                                                    |
| <b>No</b>  | INSTALL a new front camera washer relay. TEST the system for normal operation. |

**H15 CHECK THE FRONT CAMERA WASHER RELAY CONTROL CIRCUIT FOR A SHORT TO GROUND**

- Disconnect: Windshield Wiper Motor [C125](#).
- Measure the resistance between the windshield wiper motor [C125](#) Pin 12, circuit CRW34 (BU/BN), harness side.



N0138462

**Is the resistance greater than 10,000 ohms?**

|            |                                                             |
|------------|-------------------------------------------------------------|
| <b>Yes</b> | REPAIR the circuit. TEST the system for normal operation.   |
| <b>No</b>  | INSTALL a new windshield wiper motor. REFER to Wiper Motor. |